Particle Filter vs. Extended Kalman Filter Localization for a Mobile Manipulator

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This abstract will briefly describe the project and the results of the study.

I. Overview

This section will focus on introducing the localization problem briefly. Next, this section will introduce ROS2 including definitions for node and topic [1]. Then this section will describe the Gazebo simulation environment including the factory and goal points including an image of the simulation. Following, the section will introduce the hardware including the mobile platform, manipulator, and gripper along with 2D LiDAR and wheel odometry. Lastly, this section will broadly state the objectives of the study.

II. Problem Statement

This section will focus on the uncertainty in localization, kidnapping, and research questions. This section will begin with a clear description of uncertainty during localization in robotics. It will introduce the concept of kidnapping and provide examples of how this phenomenon can occur in real examples [2]. Next, this section will very briefly introduce, but not explain in detail, the localization algorithms including dead reckoning [3], extended kalman filter (EKF) [4], and particle filter (PF) [5]. At the end of this section, this section will state the hypotheses that the PF will recover localization accuracy following a kidnapping event more reliably than the EKF and that given a uniform initial pose prior, the EKF will achieve localization accuracy comparable to or better than the PF with lower computational cost.

III. Study Approach

This section will focus on the algorithm implementations, ROS2 data flow, metrics, and experiment design. It will detail the localization algorithms for dead reckoning [3], extended kalman filter (EKF) [4], and particle filter (PF) [5]. It will describe scan matching, including the iterative closest point (ICP) algorithm [6], as well as, details on the SVD-based rigid transformation [7]. Furthermore, this section will detail the design of Monte Carlo Localization (MCL) algorithm for the PF [8]. If time permits, this section will describe a hybrid algorithm in which the PF detects potential kidnapping events and supports EKF recovery by providing a pose estimate and an uncertainty estimate derived from the particle distribution.

The study approach section will continue with ROS2 node graph and data flow focusing on the inputs and outputs for the localization algorithms. It will detail where data is being produced, where noise is being injected, and the plans for storing the data for analysis. Next, this section will detail the metrics that will be collected at each time step including position error, heading error, and compute cost. Finally, this section will detail the experimental design for test cases including a uniform prior, mid-run kidnapping, the sensitivity analyses on varying particle count, injected odometry noise, and injected LiDAR noise.

IV. Results

This section will present the results of the study. It will graph the localization error over time, position RMSE in meters, heading RMSE in radians, and compute cost in seconds for the three algorithms (or four if I finish the hybrid algorithm) with 95% confidence intervals. A second set of graphs will be made for results in the kidnapping studies. Additionally, this section will provide the findings for the sensitivity analyses described in the study approach including varying particle count, injected odometry noise, and LiDAR noise.

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V. Conclusion

The final section will restate the problem statement. It will briefly overview the study approach and remark on the results and their implications. This section will include notes on limitations of the study and the conclusion will end with a discussion on future work with a focus on what would need to change to apply these algorithms to a sloped map.

References

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